



Proactive route guidance to avoid congestion



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ABSTRACT

We propose a proactive route guidance approach that integrates a system perspective: minimizing congestion, and a user perspective: minimizing travel inconvenience. The approach assigns paths to users so as to minimize congestion while not increasing their travel inconvenience too much. A maximum level of travel inconvenience is ensured and a certain level of fairness is maintained by limiting the set of considered paths for each Origin-Destination pair to those whose relative difference with respect to the shortest (least-duration) path, called travel inconvenience, is below a given threshold. The approach hierarchically minimizes the maximum arc utilization and the weighted average experienced travel inconvenience. Minimizing the maximum arc utilization in the network, i.e., the ratio of the number of vehicles entering an arc per time unit and the maximum number of vehicles per time unit at which vehicles can enter the arc and experience no slowdown due to congestion effects, is a system-oriented objective, while minimizing the weighted average experienced travel inconvenience, i.e., the average travel inconvenience over all eligible paths weighted by the number of vehicles per time unit that traverse the path, is a user-oriented objective. By design, to ensure computational efficiency, the approach only solves linear programming models. In a computational study using benchmark instances reflecting a road infrastructure encountered in many cities, we analyze, for different levels of maximum travel inconvenience and, the minimum maximum arc utilization and the weighted average experienced travel inconvenience. We find that accepting relatively small levels of maximum travel inconvenience can result in a significant reduction, or avoiding, of congestion.

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1. Introduction

The fraction of the population living in urban areas continues to grow. As a consequence, traffic in urban areas is increasing and the inability to significantly increase road network infrastructure is making the issue of traffic control and coordination more and more relevant and pressing. Congestion is a common phenomenon experienced in cities and towns around the world, and causes delays, stress, and pollution. The negative impact of congestion on the economy, the society, the environment, and on people's health is enormous. Government, industry, and private citizens are all interested in ways to reduce the negative externalities of transportation. Technology has always been an integral part of attempts to alleviate congestion, but recent and anticipated technological and automotive advances offer enormous and exciting new opportunities.

Traditionally, traffic information has been communicated to drivers via radio or by means of Variable Message Signs. The drawback of these systems is that the information being communicated is the same for all drivers and, as such, has only

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limited value in globally coordinating traffic. The most common modern in-vehicle device aimed at helping drivers guide a vehicle in a road network is a car navigation system based on a digitalized road network map and a global positioning system (GPS) aerial. The GPS aerial allows the vehicle to be localized on the map, and embedded optimization software allows the selection of the best route to the destination. Based on the available information on the status of the road network, the navigation system may provide an optimal route to the destination with respect to the user's preference, which can be the shortest path in terms of distance or travel time, or the least expensive path in terms of fuel consumption or, even, emissions produced. Challenges (and frustrations) occur when the road utilization on the route proposed by the navigation system exceeds its capacity and congestion occurs. In fact, and especially during peak hours, congestion often occurs because the paths of many vehicles traverse the same sections of the road network. Recently, navigation systems have been integrated with real-time traffic data acquisition systems that allow detection of traffic jams and/or road interruptions and offer the potential to reroute the drivers to different paths. Unfortunately, these systems typically do not (yet) consider the system-wide impact of the directions they provide to the drivers. The navigation devices, again, provide drivers with the same information and route guidance, which, in many cases, simply shifts the congestion to other parts of the road network.

The drawbacks of user-optimal paths, which result in a user equilibrium state of the traffic network, compared to a system-optimal set of user paths have long been investigated and understood (see, for example, [Mahmassani and Peeta \(1993\)](#)). It is also well-known that, with a system-optimal set of user paths, some users may end up being assigned to paths that are much longer, in distance or time, than the shortest possible path between their origin and destination. This unfairness, among others, results in users not following route guidance, especially when suggested routes deviate substantially from a user's preference and are (expected to take) much longer.

However, technological and automotive advances may change the situation favorably. The anticipated introduction of autonomous or self-driving vehicles may drastically alter the landscape. Massive adoption of self-driving vehicles will have several benefits in terms of congestion. First and foremost, because the safe separation distance between two self-driving vehicles will be much smaller, their introduction will implicitly alter the capacity of the road network. Secondly, drivers will get used to trusting their vehicle to get them from their origin to their destination and, as a consequence, will be more likely to accept a route that deviates from the preferred (shortest) route. The latter, forms the motivation and underpinning of our research. In particular, we focus on an environment in which a specific origin-destination path can be assigned to each (self-driving) vehicle and in which that vehicle will follow the assigned path. (We will conduct computational experiments to study the sensitivity to the 100% compliance assumption.)

We propose a centralized proactive route guidance approach that integrates a system perspective, focused on reducing, ideally avoiding, congestion, and a user perspective, focused on minimizing the experienced travel inconvenience. The goal is to reduce or avoid congestion without causing an excessive increase in the length (or duration) of the paths traveled by individuals, when compared to the shortest (least-duration) paths between their origin and destination.

Our starting point is a road network and an Origin-Destination (OD) matrix specifying the number of trips that are estimated to take place between each origin and each destination. The problem of forecasting traffic on a road network has been well studied (see, for example, [Ben-Akiva and Lerman \(1985\)](#); [Florian and Hearn \(1999\)](#); [Sheffi \(1985\)](#), and more recently [de Dios Ortuzar and Willumsen, 2011](#)) and traditionally has been modeled in four steps: trip generation, trip distribution, mode choice, and traffic assignment. In particular, the zonal interchange analysis of trip distribution provides the so called Origin-Destination (OD) matrix, that is the matrix that provides for each OD pair the number of trips with the same origin and destination. Starting from [Merchant and Nemhauser \(1978\)](#), researchers have also studied a dynamic traffic assignment problem which presents additional challenges (see [Papageorgiou, 1990](#); [Peeta and Ziliaskopoulos, 2001](#) and more recently [Ben-Akiva et al., 2012](#)).

The time period of interest is the rush hour, which in large cities may last a few hours, and in which, as [Sheffi \(1985\)](#) points out, traffic often exhibits a steady-state behavior. We assume that the arcs of the road network are characterized by a capacity representing the maximum number of vehicles per time unit at which vehicles can enter the arc and experience no slowdown due to congestion effects. The proposed approach will assign paths to drivers so as to minimize congestion while not increasing their experienced travel inconvenience too much. A maximum level of travel inconvenience is ensured and a certain level of fairness is maintained by limiting the set of considered paths for each Origin-Destination pair to those whose relative difference with respect to the shortest (least-duration) path, called travel inconvenience, is below a given threshold.

An important feature of the approach, and a critical design choice, is that only linear programs are solved. To have any potential practical value, a route guidance approach has to be computationally efficient. Computational efficiency has prompted us to restrict ourselves to the use of linear programming models (even if that would mean sacrificing some accuracy in our modeling choices).

The distribution of traffic is evaluated by two measures: the minimum maximum arc utilization in the network (a system perspective) and the weighted average experienced travel inconvenience (a user perspective). Arc utilization, i.e., the ratio of the number of vehicles entering an arc per time unit and its capacity, is used as an arc congestion measure. The weighted average experienced travel inconvenience averages the experienced travel inconvenience over all possible paths weighted by the number of vehicles that enter the path per time unit. The weighted average experienced travel inconvenience is minimized under the constraint that the minimum maximum arc utilization does not exceed a given limit: the minimum maximum arc utilization achievable if it is greater than one, or one if it is smaller than one (by definition no slowdown due to congestion effects occurs when the minimum maximum arc utilization is one).

The linear structure of the models allows us to derive theoretical properties. We will show in particular that for the problem of minimizing the maximum arc utilization, results analogous to those well known for the maximum flow problem, such as the max flow-min cut theorem, hold.

An extensive and comprehensive computational study demonstrates that in many settings relatively small values of the maximum travel inconvenience lead to a minimum maximum arc utilization less than or equal to one, i.e., avoidance of congestion effects. The computational tests are carried out using randomly generated benchmark instances that represent characteristic features of the most common circular-shaped city road networks. As mentioned before, our assumption is that (self-driving) vehicles follow their assigned path, i.e., a 100% compliance rate. Our investigation shows, among others, that compliance is critical; it becomes much more difficult to reduce or avoid congestion when the recommended paths are not followed.

Finding a system-optimal traffic distribution that ensures a certain level of fairness is also considered by Jahn et al. (2005). They also limit the set of paths for an OD pair to limit the travel inconvenience. However, they model the arc travel time as a function of the number of vehicles on that arc (using a widely adopted non-linear increasing function). The model assigns paths to users with the objective of minimizing the total user travel time. The model is non-linear and a column generation solution method is proposed and tested on real road networks. The approach we propose is substantially different as it seeks to avoid congestion by minimizing the maximum arc utilization. We find the minimum congestion level first and then minimize the user travel inconvenience.

A few other related papers have been published. In Sen et al. (2001), a static multi-objective approach seeking to minimize the mean travel time cost and the travel time variance is presented, while in Lujak et al. (2015) a centralized path assignment model is proposed in which the objective is to minimize the geometric path duration mean for all drivers in the system. Centrally controlled traffic systems are often encountered in large warehouses where automated guided vehicles have to be routed (see Bartlett et al., 2014). In Kaspi and Tanchoco (1990), an integer programming model is proposed which seeks to minimize the weighted path length of a set of automated guided vehicles.

The remainder of the paper is organized as follows. In Section 2, the proactive route guidance setting considered is formally introduced, the path-based linear programming models comprising our proactive route guidance approach are given, as well as an algorithm to generate the sets of eligible paths, and supporting theoretical results are presented. In Section 3, we discuss the generation of road networks on which the approach is tested, discuss detailed results for a specific instance, and average results for the complete set of instances. Finally, some conclusions are drawn in Section 4.

2. Proactive route guidance

The basic idea of the system-optimal approach we propose is to assign paths other than the shortest (least-duration) path to vehicles in order to reduce, and possibly avoid, congestion in the road network, but to do so in a way that minimizes the inconvenience experienced by the drivers, in part by only considering alternative paths whose relative increase with respect to the shortest path is within a pre-defined limit. That is, we impose a limit on user travel inconvenience, in terms of the maximum allowed increase relative to the shortest path, which we denote by τ and refer to as the *maximum travel inconvenience*.

More specifically, for each OD pair, we generate a set of eligible paths, which are those Origin-Destination paths with a travel inconvenience that is no more than the specified limit τ . Furthermore, we assume that, for the time period of interest, the demand associated with an OD pair is specified in terms of the number of vehicles entering the network at the origin per time unit. The basic premise of the approach is that the congestion of the road network depends on the utilization of its arcs, where the utilization of an arc is the ratio of the number of vehicles entering an arc per time unit and the maximum number of vehicles per time unit at which vehicles can enter the arc and experience no slowdown due to congestion effects. When the flow per time unit on an arc exceeds its capacity, i.e., the arc utilization is greater than 1, the arc is said to be *congested*. We say that the road network is congested if at least one of its arcs is congested.

Given a maximum travel inconvenience, and, thus, a set of eligible paths for each OD pair, the approach obtains a system-optimal distribution of traffic using a hierarchical approach. First, a linear programming model is used to minimize the maximum arc utilization, i.e., the level of congestion. If the minimum maximum arc utilization exceeds one, then congestion in the road network is unavoidable for the imposed maximum travel inconvenience; to reduce the level of congestion even further, longer, less convenient paths need to be allowed. On the other hand, when the minimum maximum arc utilization is less than or equal to one, congestion can be avoided with the imposed maximum travel inconvenience. In fact, there is no reason to seek a minimum maximum arc utilization below one, because it can only come at the expense of unnecessary user travel inconvenience. Second, for a given minimum maximum arc utilization, another linear programming model is used to minimize the weighted average experienced travel inconvenience subject to the constraint that the maximum arc utilization does not exceed the minimum possible value computed using the previous model (or one if this value is less than one).

Our approach focuses on a system that exhibits a steady-state behavior and seeks to determine the minimum maximum travel inconvenience that allows elimination of congestion in the system by proactive route guidance, if such a maximum travel inconvenience exists. It is possible, of course, that the optimization reveals that such a maximum travel inconvenience does not exist, due to a combination of network infrastructure characteristics, e.g., a small number of bridges and/or tunnels connecting different parts of the city, and demand characteristics, e.g., extremely high demand.

Table 1
Notation.

Problem notation	
Sets	
V	set of vertices
A	set of arcs
C	set of OD pairs
K_c^τ	set of eligible paths for $c \in C$ with maximum travel inconvenience τ
Parameters	
u_{ij}	capacity of arc $(i, j) \in A$
a_{ij}^{kc}	1 if path $k \in K_c^\tau$ contains arc $(i, j) \in A$, 0 otherwise
SP_c	length of the shortest path for OD pair $c \in C$
l_c^k	length of path $k \in K_c^\tau$
τ_c^k	relative length increment of path $k \in K_c^\tau$, with respect to SP_c : $\tau_c^k = \frac{l_c^k - SP_c}{SP_c}$
d_c	flow rate for OD pair $c \in C$
D	total flow rate over all OD pairs: $D = \sum_{c \in C} d_c$
α	compliance rate
Decision variables	
y_c^k	flow rate of OD pair $c \in C$ routed on path $k \in K_c^\tau$
Auxiliary variables	
x_{ij}	total flow rate entering arc $(i, j) \in A$: $x_{ij} = \sum_{c \in C} \sum_{k \in K_c^\tau} a_{ij}^{kc} y_c^k$
ρ	upper bound on arc utilization: $\rho \geq \frac{x_{ij}}{u_{ij}} \quad \forall (i, j) \in A$
f_{ij}^c	flow rate of the OD pair c traversing arc $(i, j) \in A$

In such situations, more elaborate linear programming models are needed. In [Section 4](#), we briefly discuss this possible extension.

Next, we present the two path-based linear programming models used in the approach, followed by the algorithm for generating eligible paths and supporting theoretical results.

2.1. Optimization models

We consider a directed network $G = (V, A)$, where V represents the set of vertices and $A \subseteq V \times V$ represents the set of arcs. Each arc $(i, j) \in A$ represents a road segment on which vehicles can travel. The length of arc $(i, j) \in A$, depending on the user's preferences, may represent the traveling time or the length in space, and it will be denoted by l_{ij} . The capacity of each arc $(i, j) \in A$ is denoted by u_{ij} and represents the maximum rate (number of vehicles per time unit) that can enter an arc (i, j) at the average speed. We consider a set C of OD pairs. An OD pair $c \in C$ is described by its origin $O_c \in V$, its destination $D_c \in V$ and a flow rate (number of vehicles per time unit) d_c , representing the rate of vehicles traveling from O_c to D_c . We let $D = \sum_{c \in C} d_c$ be the total flow rate in the network.

Let τ be the maximum travel inconvenience. A set of paths K_c^τ is associated with each OD pair c and contains all paths that are not longer than $(1 + \tau)SP_c$, where SP_c is the shortest path from the origin O_c to the destination D_c . The travel inconvenience τ_c^k associated with path $k \in K_c^\tau$, $c \in C$, is defined as the relative increase of path k with respect to the shortest (least-duration) path SP_c . Denoting by l_c^k the length of path k for OD pair c , we have $\tau_c^k = \frac{l_c^k - SP_c}{SP_c}$.

The optimization problem formulations use an indicator a_{ij}^{kc} , which takes value 1 if path $k \in K_c^\tau$ contains arc $(i, j) \in A$, and 0 otherwise. The critical parameter ρ_τ^* represents the minimum possible value of the maximum arc utilization. This value is computed by a linear programming model, referred to as the *congestion model* and defined below. The decision variables y_c^k represent the flow rate (number of vehicles per time unit) of OD pair $c \in C$ routed on path $k \in K_c^\tau$. The auxiliary variables x_{ij} and ρ represent the total flow rate (number of vehicles per time unit) traveling on arc $(i, j) \in A$ and an upper bound on the maximum arc utilization, respectively. The *inconvenience model* is a linear programming model seeking to minimize the weighted average experienced travel inconvenience over all OD pairs, defined as $\frac{1}{D} \sum_{c \in C} \sum_{k \in K_c^\tau} \tau_c^k y_c^k$. A summary of the sets, parameters and variables is given in [Table 1](#).

The inconvenience model

$$\begin{aligned} \min \quad & \frac{1}{D} \sum_{c \in C} \sum_{k \in K_c^\tau} \tau_c^k y_c^k \\ & \rho \leq \max(1, \rho_\tau^*) \end{aligned} \tag{1}$$

$$\frac{x_{ij}}{u_{ij}} \leq \rho \quad \forall (i, j) \in A \tag{2}$$

$$x_{ij} = \sum_{c \in C} \sum_{k \in K_c^\tau} a_{ij}^{kc} y_c^k \quad \forall (i, j) \in A \tag{3}$$

$$d_c = \sum_{k \in K_c^\tau} y_c^k \quad \forall c \in C \quad (4)$$

$$\rho \geq 0 \quad (5)$$

$$x_{ij} \geq 0 \quad \forall (i, j) \in A \quad (6)$$

$$y_c^k \geq 0 \quad \forall c \in C \quad \forall k \in K_c^\tau. \quad (7)$$

Constraint (1) guarantees that the maximum arc utilization is minimized in case congestion is unavoidable ($\rho_\tau^* > 1$), and ensures that the flow rate on any arc is less than or equal to the arc's capacity, otherwise ($\rho_\tau^* \leq 1$). Constraints (2) bound the maximum arc utilization, i.e., $\max_{(i,j) \in A} \{x_{ij}/u_{ij}\} \leq \rho$ and constraints (3) set the flow rate on arc (i, j) . Constraints (4) ensure that the required flow rate d_c of an OD pair $c \in C$ is routed on (a subset of) its eligible paths. Finally, constraints (5) - (7) define the domains of the decision variables. Although the natural domain of variables y_c^k and x_{ij} is the set of non-negative integers, it is reasonable to relax it to the set of real numbers as long as the values d_c are large. Note that y_c^k/d_c can be interpreted as the fraction of the flow rate d_c that is routed on path $k \in K_c^\tau$.

The minimum maximum arc utilization ρ_τ^* required in constraint (1) is computed with the following linear programming model.

The congestion model

$$\begin{aligned} \rho_\tau^* \equiv \min \quad & \rho \\ \text{s.t.} \quad & (2)-(7) \end{aligned}$$

The congestion model focuses purely on congestion, i.e., on minimizing the maximum arc utilization, without taking into account the impact on the drivers. On the other hand, the inconvenience model without constraints (1) - (2) focuses purely on user travel inconvenience without taking into account the impact on congestion. For this reason, a hierarchical approach is used. We first focus on congestion and then focus on user travel inconvenience.

The following remarks provide basic relations linking the maximum travel inconvenience τ , the path set K_c^τ , and the minimum value of the maximum arc utilization ρ_τ^* .

Remark 1. Let $K_c^{\tau_1}$, $K_c^{\tau_2}$ be the sets of eligible paths for OD pair $c \in C$ for τ_1 and τ_2 , respectively. If $\tau_1 \leq \tau_2$, then $K_c^{\tau_1} \subseteq K_c^{\tau_2}$.

Remark 2. Let $K_c^{\tau_1}$, $K_c^{\tau_2}$ be the sets of eligible paths for OD pair $c \in C$ for τ_1 and τ_2 , respectively. If $K_c^{\tau_1} = K_c^{\tau_2} \quad \forall c \in C$, then the minimum maximum arc utilization is the same for the two settings.

Remark 3. Let $\rho_{\tau_1}^*$, $\rho_{\tau_2}^*$ be the optimum values of the congestion model when the maximum travel inconvenience is τ_1 and τ_2 , respectively. If $\tau_1 \leq \tau_2$ then $\rho_{\tau_1}^* \geq \rho_{\tau_2}^*$.

The models can be modified to take into account that not all vehicles may follow the recommended path. We define the *compliance rate* to be the fraction α of vehicles following the recommended path and assume that $1 - \alpha$ vehicles, instead, choose the shortest (least-duration) path. For sake of simplicity, we assume that the compliance rate is identical for all OD pairs. The compliance rate is taken into account by adding to each model the constraints

$$y_c^{SP} \geq (1 - \alpha)d_c \quad \forall c \in C. \quad (8)$$

These constraints impose that at least a fraction $1 - \alpha$ of the demand is routed along its shortest path. This fraction represents those drivers acting “selfishly”.

2.2. Generation of the eligible paths

To the best of our knowledge, whereas the problem of finding the first K shortest paths is well known (see Yen, 1971 and Eppstein, 1994), no algorithm has been published for the computation of all shortest paths between two vertices whose length does not increase the shortest path length by a given percentage. Thus, in order to compute the set K_c^τ , an ad hoc algorithm has been developed. The algorithm relies on depth-first search and consists of a main body *ConstructingEligiblePaths* (sketched in Algorithm 1) and a recursive routine *ScanVertex* (sketched in Algorithm 2).

Inputs for algorithm *ConstructingEligiblePaths* are the network (V, A) , the set of OD pairs C , and the maximum travel inconvenience τ . The algorithm first computes the shortest path matrix (the shortest path for each pair of vertices) and labels all vertices as inactive. Labels will change from inactive to active and back during computation. Next, for each OD pair $c \in C$, initializations are made and a depth-first search to find the set K_c^τ is performed by means of the recursive routine *ScanVertex*.

Routine *ScanVertex* operates as follows. The input vertex i is labeled as active (already visited) and inserted in the growing path at the current position. If the input vertex is the destination vertex the procedure records the path, labels the vertex as inactive and exits. Otherwise, the routine is recursively called on each vertex reached by any outgoing arc from the input

Algorithm 1: ConstructingEligiblePaths

```

input :  $V, A, C, \tau$ 
 $s \leftarrow \text{ComputeShortestPathMatrix}(V, A)$ 
for  $i \in V$  do
   $\text{status}(i) \leftarrow \text{INACTIVE}$ 
for  $c \in C$  do
   $\text{Path} \leftarrow \emptyset$ 
   $c\text{Path} \leftarrow 0$ 
   $l\text{Path} \leftarrow 0$ 
   $\text{ScanVertex}(O_c)$ 

```

Algorithm 2: ScanVertex

```

input :  $i$ 
global variables:  $s, \tau, \text{status}, \text{Path}, c\text{Path}, l\text{Path}$ .
 $\text{status}(i) \leftarrow \text{ACTIVE}$ 
 $\text{Path}(c\text{Path}) \leftarrow i$ 
if  $i = D_c$  then
   $\text{Record}(\text{Path})$ 
else
  for  $j \in \delta^+(i)$  do
    //  $j$  is the endpoint of an arc exiting from  $i$ 
    if  $\text{status}(j) = \text{INACTIVE}$  and  $l\text{Path} + l_{ij} + s(j, D_c) \leq (1 + \tau)SP_c$  then
       $c\text{Path} \leftarrow c\text{Path} + 1$ 
       $l\text{Path} \leftarrow l\text{Path} + l_{ij}$ 
       $\text{ScanVertex}(j)$ 
       $l\text{Path} \leftarrow l\text{Path} - l_{ij}$ 
       $c\text{Path} \leftarrow c\text{Path} - 1$ 
   $\text{status}(i) \leftarrow \text{INACTIVE}$ 

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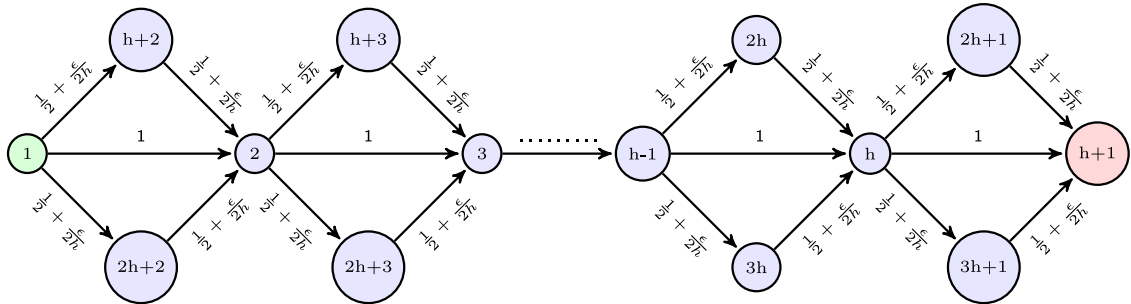


Fig. 1. An instance demonstrating the exponential complexity of the algorithm for generating the sets of eligible paths for the OD pairs.

vertex provided that the new vertex is not labeled as active (which would imply a cycle) and the length of the growing path is within the fixed threshold. The vertex count and the total length are increased and decreased before and after recursion.

The algorithm has time complexity $\mathcal{O}(q^{|V|+1})$, where q is an upper bound on the number of outgoing arcs from any vertex. The exponential complexity is due to the problem nature as explained in the following example. Consider the network, consisting of $n = 3h + 1$ vertices and with $q = 3$, shown in Fig. 1. Consider vertex 1 as O_c and vertex $h + 1$ as D_c and let $\epsilon \leq \tau$. The h arcs connecting the $h + 1$ central vertices with length 1 define the shortest path from O_c to D_c , and, thus, $SP_c = h$. The remaining arcs have length $\frac{1}{2} + \frac{\epsilon}{2h}$. All possible paths in this network from O_c to D_c belong to the set K_c^τ . The number of the possible paths is $3^h = 3^{\frac{n-1}{3}}$.

2.3. Lower bounds on the minimum maximum arc utilization

In this section we propose an arc-based formulation for the congestion model where, for each OD pair c , all possible paths from O_c to D_c are implicitly considered. The model, which we refer to as the *unconstrained congestion model*, is equivalent to the congestion model in which the maximum travel inconvenience is set to $+\infty$. Allowing all paths for all the OD pairs means that the result of the unconstrained congestion model is a lower bound for the congestion model for any value of τ .

The unconstrained congestion model

$$\rho_\infty^* \equiv \min \rho$$

$$\sum_{j \in V} f_{ij}^c - \sum_{j \in V} f_{ji}^c = 0 \quad \forall c \in C \quad \forall i \in V, \quad i \neq O_c, \quad i \neq D_c \quad (9)$$

$$\sum_{j \in V} f_{ij}^c - \sum_{j \in V} f_{ji}^c = d_c \quad \forall c \in C \quad i = O_c \quad (10)$$

$$\sum_{j \in V} f_{ij}^c - \sum_{j \in V} f_{ji}^c = -d_c \quad \forall c \in C \quad i = D_c \quad (11)$$

$$\rho \geq \frac{\sum_{c \in C} f_{ij}^c}{u_{ij}} \quad \forall c \in C \quad \forall (i, j) \in A \quad (12)$$

$$f_{ij}^c \geq 0 \quad \forall c \in C \quad \forall (i, j) \in A \quad (13)$$

$$\rho \geq 0. \quad (14)$$

Constraints (9)–(11) guarantee the flow rate conservation. The objective function together with Constraints (12) set the minimum maximum arc utilization to the minimum of the maximum value, over all arcs, of the ratio between the total flow rate and the arc capacity. Finally, Constraints (13) and (14) define the domains of the decision variables.

Remark 4. The value ρ_∞^* of the unconstrained congestion model is a lower bound for ρ_τ^* , for any value of the maximum travel inconvenience τ , i.e.,

$$\rho_\tau^* \geq \rho_\infty^* \quad \forall \tau.$$

Thus, the value ρ_∞^* is the minimum level of congestion that can be achieved when vehicles can be sent along any possible path (regardless of the experienced travel inconvenience). This value may be greater than one due to the structure of the network and high levels of the traffic. In the following, we derive lower bounds on ρ_∞^* that can be computed from the network parameters. The results we derive recall results that are well known for the maximum flow problem (see Ford and Fulkerson, 1956).

Definition 1. A cut-set for a set of OD pairs $\hat{C} \subseteq C$ is a minimal arc set $A_{\hat{C}}$ such that in the graph $G' = (V, A \setminus A_{\hat{C}})$ no path connecting O_c to D_c exists for any OD pair in $\hat{C}$. The capacity of a cut-set $A_{\hat{C}}$ is $u_{\hat{C}} = \sum_{(ij) \in A_{\hat{C}}} u_{ij}$.

Definition 2. A minimum capacity cut-set for $\hat{C}$ is a cut-set $A_{\hat{C}}^*$ with minimum capacity. We denote the capacity of $A_{\hat{C}}^*$ by $u_{\hat{C}}^*$.

Theorem 1. Let $A_{\hat{C}}^*$ be a minimum cut-set for any set of OD pairs $\hat{C} \subseteq C$, and let $u_{\hat{C}}^*$ be its capacity. Then,

$$\rho_\infty^* \geq \max_{\hat{C} \subseteq C} \frac{\sum_{c \in \hat{C}} d_c}{u_{\hat{C}}^*}.$$

Proof. Let $\tilde{G}$ be a network in which the capacities $\hat{u}_{ij}$ are the capacities u_{ij} of the original network G multiplied by the minimum maximum arc utilization ρ_∞^* , i.e. $\hat{u}_{ij} = \rho_\infty^* u_{ij}$. Let $\hat{C} \subseteq C$ be a set of OD pairs and $A_{\hat{C}}^*$ a minimum cut-set for $\hat{C}$. In $\tilde{G}$ it is possible to route the demand of the OD pairs in $\hat{C}$ without violating the capacities $\hat{u}_{ij}$. Hence, $\sum_{c \in \hat{C}} d_c \leq \sum_{(ij) \in A_{\hat{C}}^*} \hat{u}_{ij} = \sum_{(ij) \in A_{\hat{C}}^*} \rho_\infty^* u_{ij} = \rho_\infty^* \sum_{(ij) \in A_{\hat{C}}^*} u_{ij} = \rho_\infty^* u_{\hat{C}}^*$. Hence, $\frac{\sum_{c \in \hat{C}} d_c}{u_{\hat{C}}^*} \leq \rho_\infty^*$. As this inequality holds for any subset $\hat{C} \subseteq C$, the claim follows. $\square$

In the following, we show that in the case of a single OD pair equality holds, but that examples can be found with strict inequality for the case of multiple OD pairs.

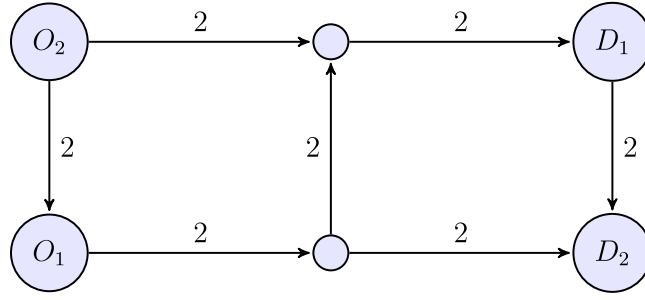


Fig. 2. Example for two OD pairs.

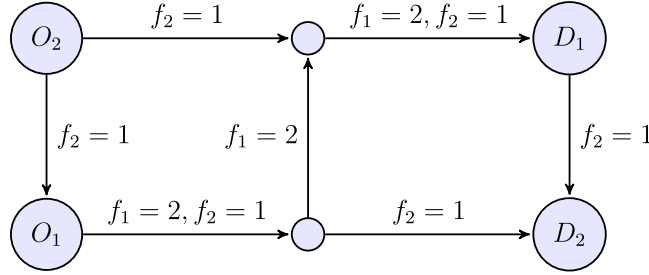


Fig. 3. Optimal flows on counterexample graph.

Theorem 2. Let $|C| = 1$ and let us denote by c the OD pair. Let A_c^* be a minimum cut-set for c with capacity u_c^* . Then, $\rho_\infty^* = \frac{d_c}{u_c^*}$.

Proof. See Appendix Appendix A. $\square$

Remark 5. Equality $\rho_\infty^* = \max_{\hat{C} \subseteq C} \frac{\sum_{c \in \hat{C}} d_c}{u_c^*}$ does not hold when there are two or more OD pairs in the network, that is when $|C| \geq 2$.

This example shows that the equality does not hold in the case of two OD pairs. In Fig. 2 all arcs have capacity u equal to 2 and each OD pair has demand equal to 2. Deriving a lower bound on the minimum maximum arc utilization means exploring all the possible subsets of OD pairs in C . The number of possible subsets in C is given by the cardinality of the power set of C , $P(C)$. The cardinality of the power set depends exponentially on the number of OD pairs contained in C , i.e. $|P(C)| = 2^n$, where n is the number of OD pairs in C .

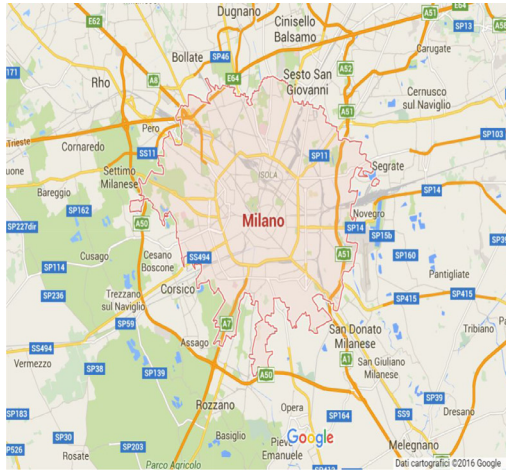
In Fig. 3 the optimal flows for each OD pair are shown. It is easy to see that the demand is completely routed. The minimum maximum arc utilization value is equal to $\rho_\infty^* = \frac{3}{2}$. Considering each subset $\hat{C} \subseteq C$ and calculating $\rho_\infty^{\hat{C}} = \frac{\sum_{c \in \hat{C}} d_c}{u_c^*}$ for each subset, we obtain that the maximum of all these values is $\max_{\hat{C} \subseteq C} \rho_\infty^{\hat{C}} = \max_{\hat{C} \subseteq C} \frac{\sum_{c \in \hat{C}} d_c}{u_c^*} = 1 < \rho_\infty^*$.

3. Computational results

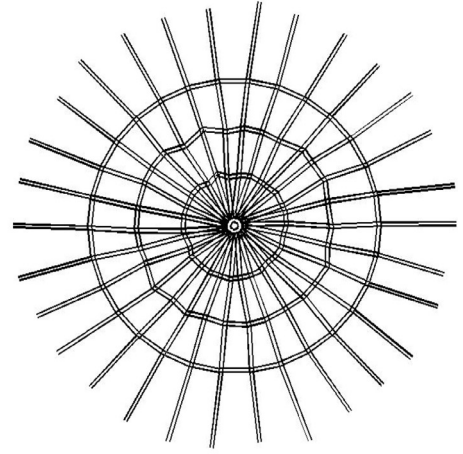
The instance generator was implemented in Java, and the optimization models were solved by CPLEX 12.6.0. The generation of the instances and the experiments were run on a Windows 64-bit computer with Intel Xeon processor E5-1650, 3.50 GHz, and 16 GB Ram. Instance generation depends on a few controls, which are described in more detail in 3.1. Considering all possible combinations of these controls, the number of different types of networks is 16. Instance generation also depends on some random factors which are not included into the control set. For example, the coordinates of the nodes in a network is randomly perturbed from their original position. For each set of values for the controls, we generate 5 random instances. The total number of generated instances is therefore 80. The instances are available on <http://or-brescia.unibs.it/instances>. The statistics collected for each instance are described in Section 3.2. Detailed results and insights for a single, specific instance are presented and discussed in Section 3.3. Section 3.4 is devoted to summary results for all instances.

3.1. Instance generation

We designed a procedure that generates instances reflecting the critical aspects of real city road networks. The procedure allows modeling of highways, bypasses, orbital roads and suburb streets, with capacities that depend on the road

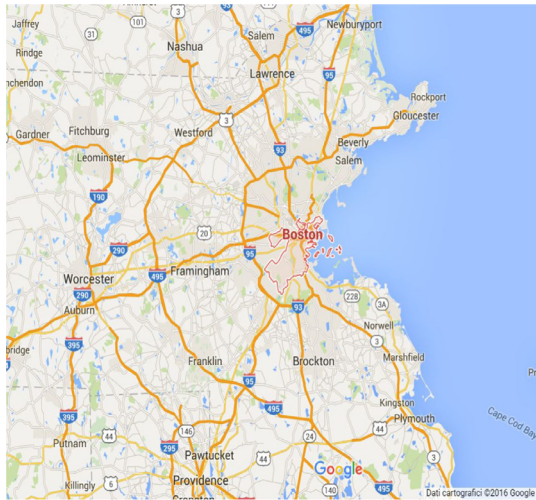


(a) Map extract

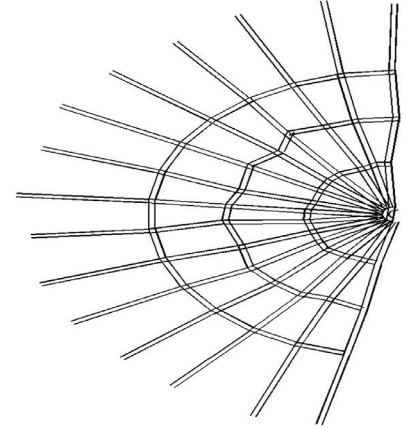


(b) City network

Fig. 4. City with a circular road infrastructure.



(a) Map extract



(b) City network

Fig. 5. City with a semi-circular road infrastructure.

importance. The procedure can generate road networks with different shapes, for example with a circular structure (see Fig. 4) or a semi-circular structure to be able to capture cities built adjacent to natural structures such as the sea or hills (see Fig. 5). However, for the computational study, we restrict ourselves to instances with a circular structure as it represents the more general case. A circular road network has an inner ring, an outer ring, and some intermediate rings. The inner ring surrounds the central area and is considered a low-speed ring which allows access to the downtown area. The outer ring is a heavy traffic ring where all traffic coming from suburbs and surrounding towns converges. Depending on the city size, there can be one or more intermediate high-speed rings. Radial roads emanate outwards from the city center. Such roads finish at locations outside the outer ring. These locations are used in our experiments to represent origins (destinations) of traffic from (to) hinterland towns or highway connection points, i.e., the main roads entering the metropolitan area. The main roads entering the city are randomly classified as highways or high capacity roads. The radial roads also link the different rings.

The instance generator has a number of parameters which allow us to generate different instances. In the following, we present groups of parameters that in combination define a major characteristic of the generated instance; we call such a group a *control*. We also name two specific sets of values of the parameters of a control to clearly identify different instance classes.

- **Control A: size**

This control determines the size of the city. We distinguish small-to-medium cities and Big metropolitan areas:

- Small-to-medium
 - * Internal radius: 2000 m
 - * External radius: 6000 m
 - * Number of rings: 2
 - * Number of radial roads: 15
 - * Number of entering main roads: 2
- Big
 - * Internal radius: 1000 m
 - * External radius: 20000 m
 - * Number of rings: 4
 - * Number of radial roads: 30
 - * Number of entering main roads: 6

- **Control B: angular width**

This control determines the city shape. The generator takes as input a city angular width β measured in degrees. We generate only circular cities:

- circular: $\beta=360$

- **Control C: attraction points**

In most cities, there are one or more attraction points, such as malls, stadiums, office parks, and schools. The generator takes as input the percentage π of nodes that represent attraction points. We distinguish oligo-centric and poli-centric cities:

- oligo-centric: $\pi = 40\%$
- poli-centric: $\pi = 80\%$

- **Control D: traffic density**

The flow for an OD pair is set by multiplying the sum over all origin outgoing arc capacities by a value σ , drawn uniform randomly from some specified interval. We distinguish off-peak and in-peak traffic densities for each network size:

- Small-to-medium
 - * off-peak: $\sigma \in [0.2; 0.4]$
 - * in-peak: $\sigma \in [0.2; 0.6]$
- Big
 - * off-peak: $\sigma \in [0.1; 0.3]$
 - * in-peak: $\sigma \in [0.3; 0.6]$

- **Control E: in-city traffic**

Most of city traffic originates in the surrounding areas, but a percentage comes from locations inside the external ring. For this reason, the generator creates a percentage, γ , of OD pairs in C with both origin and destination inside the city. We distinguish low and high in-city traffic:

- low: $\gamma=10\%$
- high: $\gamma=20\%$

The capacity of each road is computed taking into account the following parameters:

- Number of lanes,
- Average speed,
- Safe separation distance.

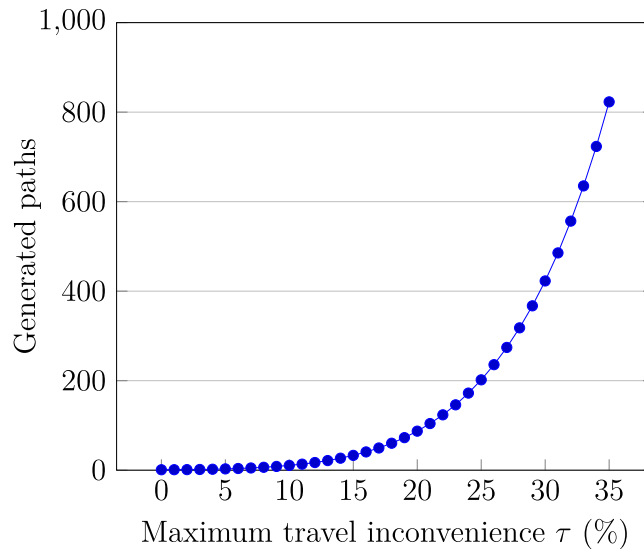
We generate instances with four different classes of roads: Highways, Main roads, Secondary Roads and Links between roads. Each class has different values of the parameters listed above. In particular, Highways have the highest number of lanes and the largest safe separation distance, while Secondary roads have the smallest number of lanes and the smallest safe separation distance. Also the average speed value differs from class to class. We note that the average speed takes into account traffic lights and other traffic regulations. For the sake of simplicity we have assumed that the speed is the same on all roads. In this case using distance or travel time as length is equivalent.

3.2. Statistics

In our experiments, we vary the value of the maximum travel inconvenience from $\tau = 0\%$ (all vehicles follow the shortest origin-destination path) to $\tau = 35\%$ (and the values of the compliance rate from 100% to 60%). We collect and compute the statistics shown in Table 2. To present large amounts of information in an easy-to-interpret and concise form, we mostly rely on graphs.

Table 2
Statistics.

Network congestion	
ρ_τ^*	The minimum maximum arc utilization ρ_τ^* obtained from the congestion model.
x_{ij}/u_{ij} classes	The fraction of arcs falling in each of the following four classes for different τ values: • unused arcs ($x_{ij}/u_{ij} = 0$) • non-congested arcs ($0 < x_{ij}/u_{ij} \leq 1$) • lightly congested arcs ($1 < x_{ij}/u_{ij} \leq 1.5$) • heavily congested arcs ($1.5 < x_{ij}/u_{ij}$).
No-congestion τ	The no-congestion travel inconvenience, i.e. the minimum value of maximum travel inconvenience needed to avoid congestion.
User experience	
Experienced inconvenience	The weighted average experienced travel inconvenience, i.e. the optimal value of the inconvenience model.
Inconvenience classes	The percentage of the demand experiencing different levels of travel inconvenience: • level A: from 0% to 5% • level B: from 5% to 10% • level C: from 10% to 15% • level D: from 15% to 20% • level E: from 20% to 25% • level F: from 25% to 30% • level G: from 30% to 35%.
OD paths	
Generated paths	Average number of generated paths per OD pair.
Selected paths	Selected paths in the optimal solution of the inconvenience model: • average number of selected paths per OD pair • maximum number of selected paths per OD pair.

**Fig. 6.** Average number of generated paths.

3.3. Detailed results and insights for a specific instance

In this section, we analyze the traffic patterns produced by the proactive route guidance approach for a single instance. We chose an instance with a large, circular, and oligo-centric city network with high in-city traffic and in-peak traffic density. In order to provide a detailed understanding of the potential benefits of adopting proactive route guidance, we solved the model with τ ranging from 0% to 35% in increments of 1%.

In Fig. 6, we show the average number of generated paths per OD pair. As expected, the average number of generated paths grows exponentially. Clearly, at higher values of the maximum travel inconvenience, there are more paths to choose from and it should be possible to reduce congestion more.

In Fig. 7, we show the minimum maximum arc utilization ρ_τ^* for the different values of the maximum travel inconvenience τ . As expected, ρ_τ^* is a non-increasing function of τ . We note that for $\tau = 26\%$ the value of ρ_τ^* drops below 1, which implies that there is no congestion in the system. Consequently, for values of the maximum travel inconvenience greater than or equal to 26%, the proactive route guidance approach minimizes the weighted average experienced travel inconvenience while ensuring that the minimum maximum arc utilization does not exceed one. (We note that the experienced

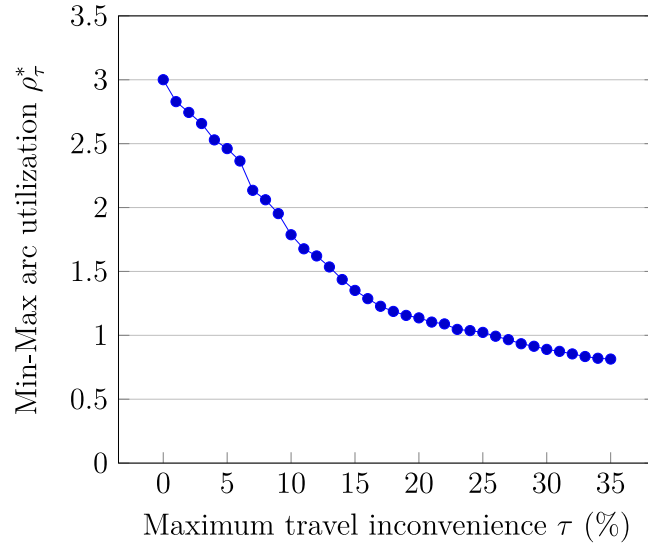


Fig. 7. Minimum maximum (Min-Max) arc utilization.

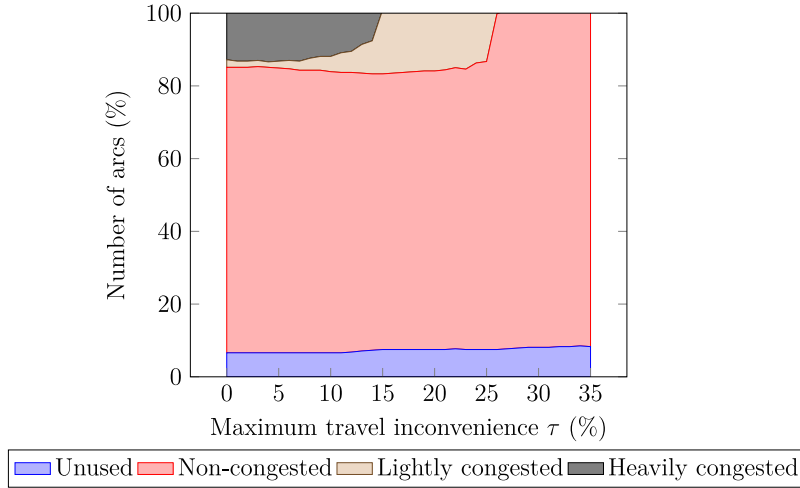


Fig. 8. Arc utilization distribution.

travel inconvenience value is realistic only when a path has no arcs with an arc utilization that exceeds one. Otherwise, the experienced travel inconvenience value represents an underestimate, as congestion will be encountered along the path.)

In Fig. 8, we show the arc utilization distribution for the different values of the maximum travel inconvenience. We see that the fraction of unused arcs in the road network remains almost steady. When τ increases from 0% to 13%, the fraction of heavily congested arcs decreases, but, at the same time, the fraction of lightly congested arcs increases and the fraction of the non-congested arcs slightly decreases. We have no heavily congested arc for $\tau \geq 14\%$, and, as previously observed, the road network becomes free of congestion when $\tau \geq 26\%$. The two seemingly abrupt transitions from $\tau = 13\%$ to $\tau = 14\%$ when the road network loses heavily congested arcs, and from $\tau = 25\%$ to $\tau = 26\%$ when the road network loses all congested arcs, are smoother than they may appear. Indeed, for $\tau = 13\%$ we have $\rho_\tau^* = 1.54$ and all the 8.3% heavily congested arcs are very close to be classified as lightly congested. For $\tau = 25\%$ we have $\rho_\tau^* = 1.02$ and all the congested arcs are very close to be classified as non-congested.

In Fig. 9, we show the weighted average experienced travel inconvenience for maximum travel inconvenience values greater than or equal to 26%, i.e., values for which congestion can be eliminated. As expected, we see that the weighted average experienced travel inconvenience decreases, because additional paths are exploited to reduce the weighted average experienced travel inconvenience rather than to reduce the maximum arc utilization.

In Fig. 10, we show, for $\tau = 26\%$, the number of OD pairs experiencing a certain level of travel inconvenience. Importantly, we see that for almost half of the OD pairs travel still occurs along the shortest path and that only for a relatively small number of OD pairs travel inconvenience exceeds 15%.

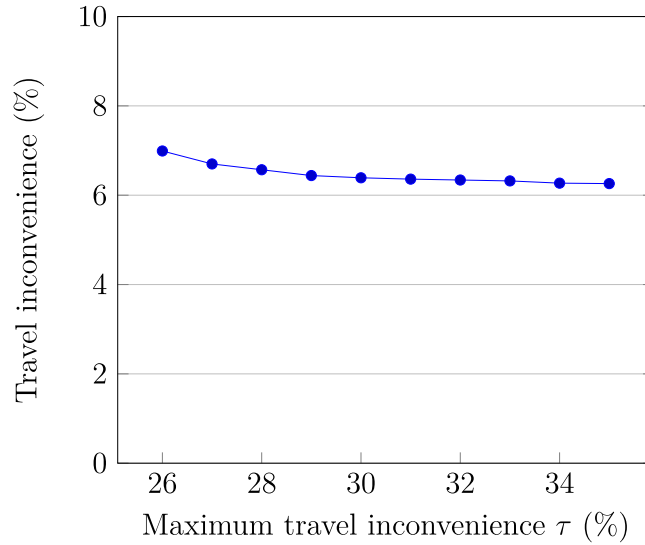


Fig. 9. Weighted average experienced travel inconvenience.

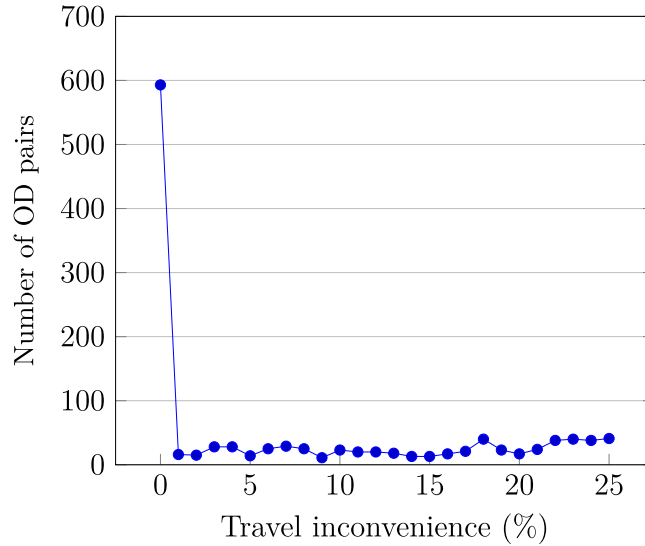


Fig. 10. Experienced travel inconvenience at $\tau = 26\%$.

The pattern shown in Fig. 10 is also seen in Fig. 11, where we show the fraction of OD pairs that is experiencing a certain travel inconvenience level (A through G) for values of $\tau \geq 26\%$. We see that most vehicles are sent along paths with little or no travel inconvenience (level A) and that only a small number of vehicles are sent along paths with a high travel inconvenience (levels E through G).

In Fig. 12, we show the average and the maximum number of selected paths for an OD pair. We observe that the maximum number of paths selected for an OD pair is surprisingly small. For all but one of the maximum travel inconvenience values, the number of selected paths is less than or equal to three. This implies that vehicles with the same origin and destination are assigned to only a small number of different paths. (This will facilitate developing schemes that assign paths in a fair way over time, e.g., a round-robin assignment scheme.) We also observe that for all maximum travel inconvenience values, for most of the OD pairs, the vehicles are routed along a single path, as the average number of paths selected for an OD pair is very close to 1. However, we note that when a single path is selected for an OD pair, this path does not necessarily have to be the shortest path and that, in fact, in many cases it is not.

Even though we are focusing on an environment in which a specific origin-destination path can be assigned to each vehicle and in which that vehicle will follow the assigned path, it is informative to examine what happens when the 100% compliance assumption does not hold. Fig. 13 shows the minimum maximum arc utilization ρ_τ^* that can be achieved as a function of maximum travel inconvenience for different compliance rates. We see that for $\alpha = 90\%$ congestion can still be

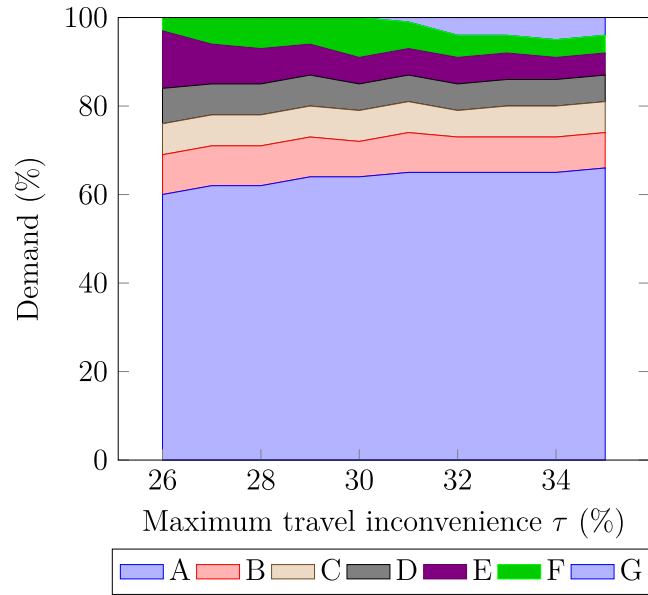


Fig. 11. Experienced travel inconvenience distribution for various levels of maximum travel inconvenience.

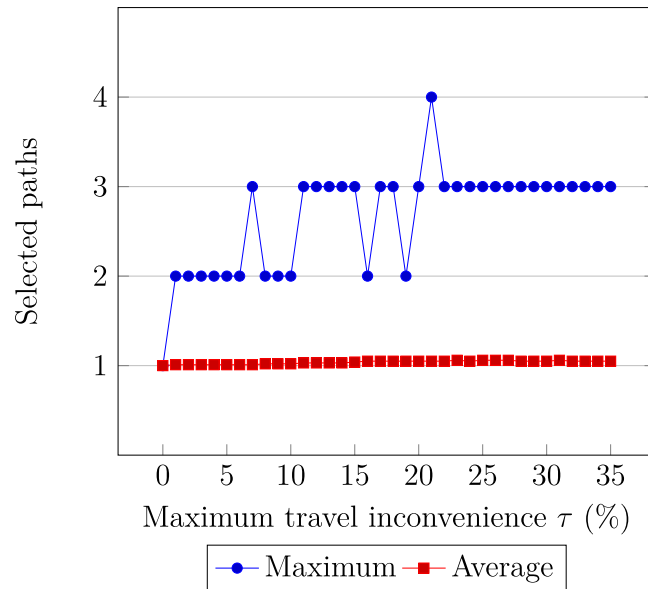


Fig. 12. Selected paths.

avoided as the minimum maximum arc utilization falls below 1 at $\tau = 30\%$, but that for lower levels of compliance, this is no longer possible.

3.4. Summary results for all instances

In this section, we present summary statistics for groups of instances. In Figs. 14–16, we group instances that require the same maximum travel inconvenience to be able to reach the no-congestion state. In the final two charts, i.e., Figs. 17–18, we look at all instances, but for specific maximum travel inconvenience values.

In Fig. 14, we show for each value of the maximum travel inconvenience the number of instances for which it is possible to reach a no-congestion state (distinguishing in-peak and off-peak traffic densities). We observe that with off-peak traffic density at $\tau = 22\%$ it is possible to reach a no-congestion state for all 40 instances, but that with in-peak traffic density even at $\tau = 35\%$ it is not possible to reach the no-congestion state for two instances. However, the value of minimum maximum arc utilization for these two instance is very close to one, and a no-congestion state is attained at $\tau = 36\%$.

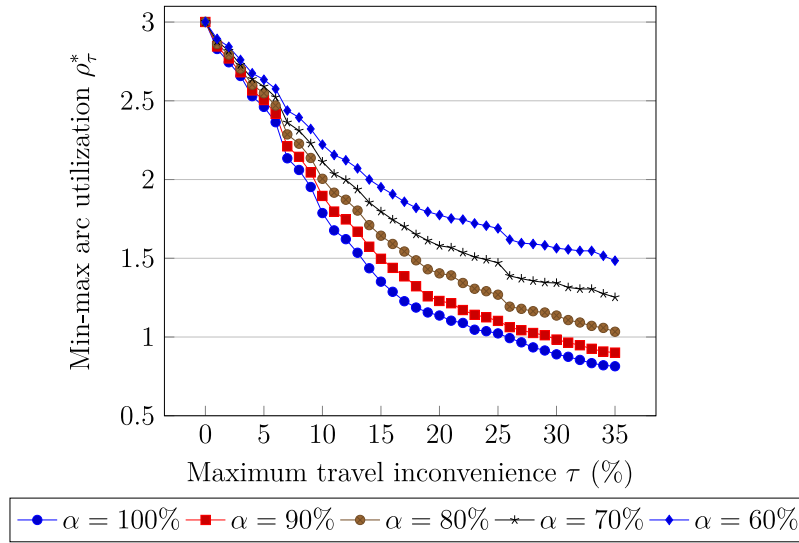


Fig. 13. Impact of the compliance rate on the minimum maximum (min-max) arc utilization.

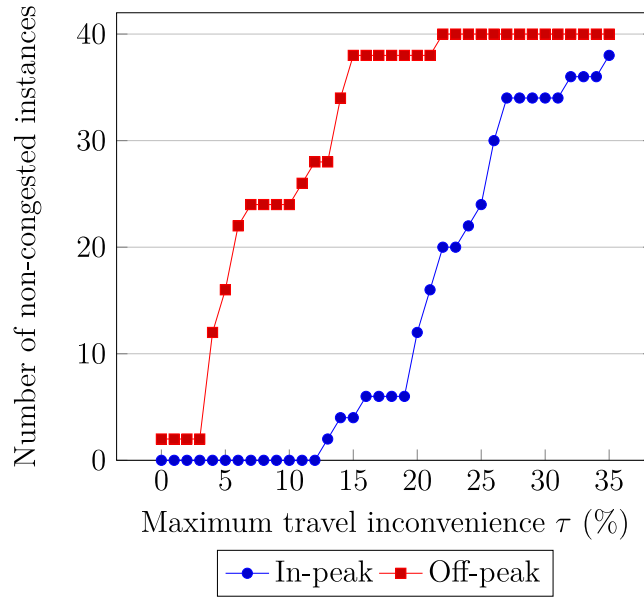


Fig. 14. Number of instances in a no-congestion state for different values of τ (%).

In Figs. 15 and 16, we consider particular statistics and average this statistic over all instances that reach the no-congestion state at a particular maximum travel inconvenience value. In Fig. 15, we show the average and the maximum number of selected paths. We observe that the results suggest that when a higher maximum travel inconvenience is required to reach the no-congestion state, more OD pairs will be assigned more than one path as both the average of the maximum number of selected paths and the average of the average number of selected paths are (slightly) higher. In Fig. 16, we show the weighted average experienced travel inconvenience. We observe that, as expected, the weighted average experienced travel inconvenience is higher when it is more difficult to reach a no-congestion state, but also that the weighted average experienced travel inconvenience remains relatively small even for instances where it is challenging to reach a no-congestion state.

In Fig. 17, we show the average minimum maximum arc utilization over all instances with in-peak and off-peak traffic densities. As expected, we see that the minimum maximum arc utilization is (significantly) smaller when the traffic density is smaller, as fewer vehicles have to be accommodated in the same road network. We see that with in-peak traffic density it is possible (on average) to reach a no-congestion state at $\tau = 23\%$. On the other hand, with off-peak traffic density, a no-congestion state is reached (on average) at $\tau = 8\%$.

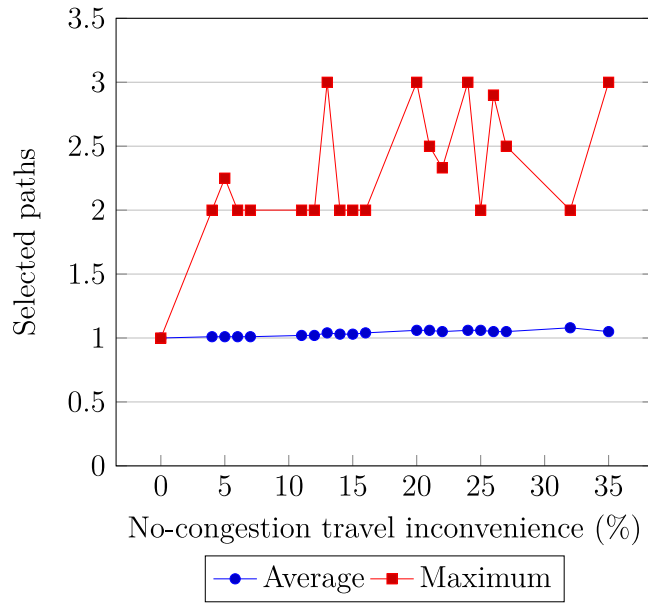


Fig. 15. Number of paths chosen at the no-congestion travel inconvenience τ (%).

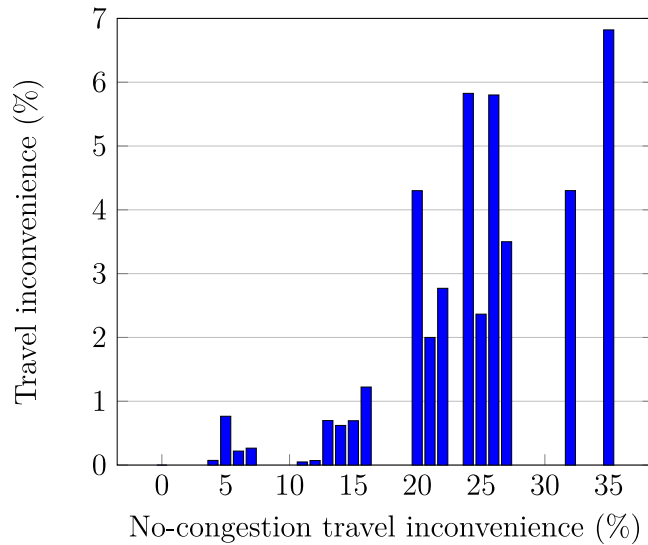


Fig. 16. Weighted average experienced travel inconvenience at the no-congestion travel inconvenience value (%).

In Fig. 18, we show the average arc utilization distribution. The average number of unused arcs remains almost the same while the average number of non-congested arcs steadily increases with increasing values of maximum travel inconvenience. At $\tau = 35\%$, the proactive route guidance approach is able to eliminate heavily congested arcs in all instances and reduce the average number of lightly congested arcs to less than 0.5%. This congestion “residue” is due to two instances that remain lightly congested at $\tau = 35\%$.

4. Final remarks and future research

In this paper, we have conducted a computational study of the potential of proactive route guidance to reduce or avoid congestion in an environment in which a specific origin-destination path can be assigned to each vehicle in the system and in which that vehicle will follow the assigned path. We believe that the advent of autonomous (or self-driving) vehicles makes this a (much more) realistic assumption.

Our proposed approach has an important (computational) advantage over previously proposed approaches: it relies only on linear programming. As such, it is more likely to scale and be usable in real-world settings. However, much needs to happen to make that a reality. One important step in that direction, and one that we are currently pursuing, is the dynamic

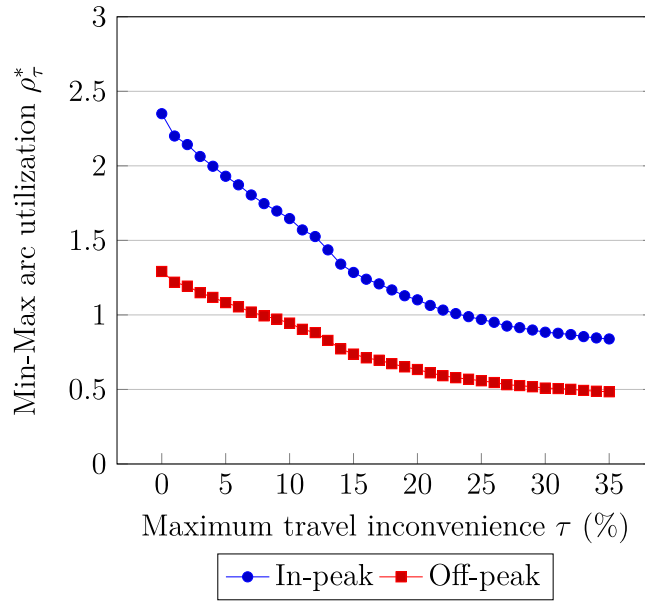


Fig. 17. Minimum maximum (Min-Max) arc utilization.

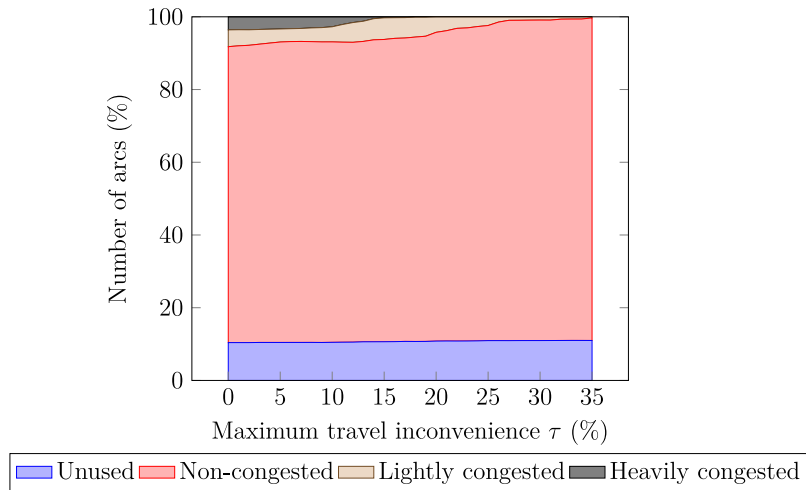


Fig. 18. Arc utilization distribution.

generation of paths (as opposed to generating the set of eligible paths upfront). Given the exponential growth of the sets of eligible paths when the maximum travel inconvenience increases, this will be critical.

Another critically important extension is incorporating traffic-dependent arc travel times, which can be done efficiently by employing piecewise linear approximations of these functions. This will allow more accurate modeling of travel times along paths that contain one or more congested arcs, i.e., arcs with a utilization greater than one, which is especially important in situations where it is not possible to reach a no-congestion state.

A final extension, but one that is more involved, considers varying demand rates over time (instead of assuming steady-state behavior), which will allow more accurate modeling of the traffic dynamics.

Acknowledgments

The authors would like to thank the anonymous reviewers for their valuable comments and suggestions to improve the quality of the paper.

Appendix A. The one OD pair case proof

Theorem 3. Let $|C| = 1$ and let us denote by c the OD pair. Let A_c^* be a minimum cut-set for c with capacity u_c^* . Then, $\rho_\infty^* = \frac{d_c}{u_c^*}$.

Proof. Let $\hat{\rho} = \frac{d_c}{u_c^*}$. We prove that $\rho_\infty^* = \hat{\rho}$ in two steps, showing that:

1. There exists a flow $\hat{f}_{ij}$, $\forall (i, j) \in A$, such that:
 - d_c flows from O_c to D_c ,
 - $\hat{f}_{ij} \leq \hat{\rho}$, $\forall (i, j) \in A$,
 - $\hat{f}_{ij} = \hat{\rho}$ for some $(i, j) \in A$.
2. There does not exist a flow g_{ij} , $\forall (i, j) \in A$, such that:
 - d_c flows from O_c to D_c ,
 - $\frac{g_{ij}}{u_{ij}} < \hat{\rho}$, $\forall (i, j) \in A$.

1. We solve the maximum flow problem on the graph G for the OD pair c . Let $\delta^+(O_c)$ be the set of the successors of O_c . From the max flow-min cut theorem, the maximum flow $F_{\max}$ is equal to the minimum cut-set capacity u_c^* . Flows f_{ij} in the optimal solution of the maximum flow problem are such that:

- $f_{ij} \leq u_{ij}$, $\forall (i, j) \in A$,
- $f_{ij} = u_{ij}$, $\forall (i, j)$ such that $(ij) \in A_c^*$.

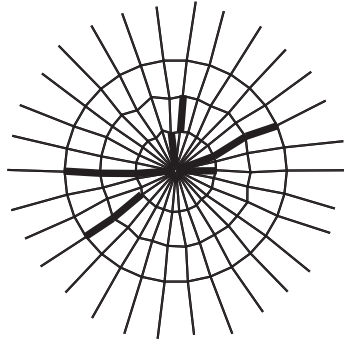
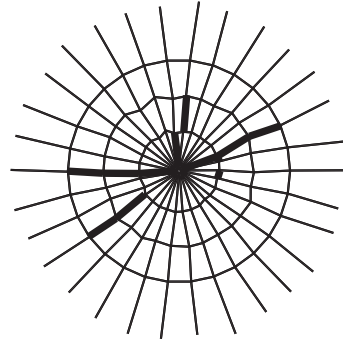
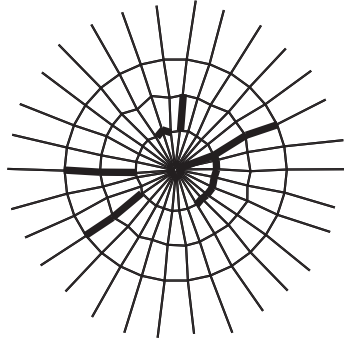
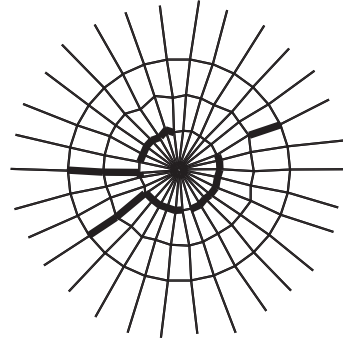
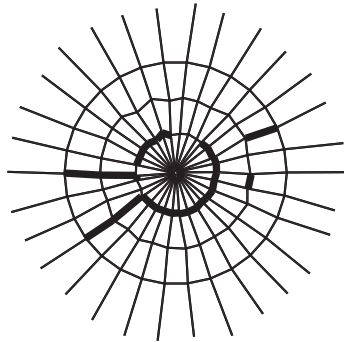
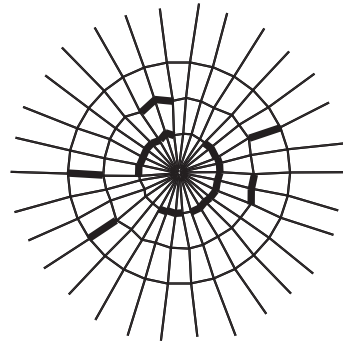
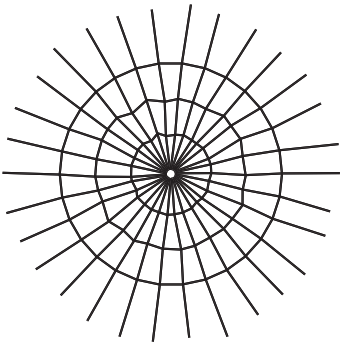
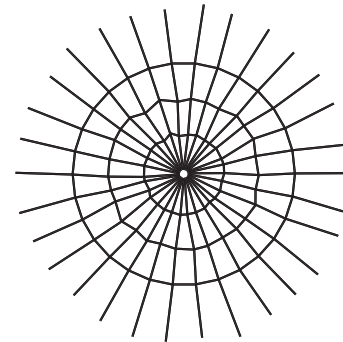
We construct new flows $\hat{f}_{ij} = f_{ij}\hat{\rho}$, $\forall (i, j) \in A$, and obtain:

- $\sum_{j \in \delta^+(O_c)} \hat{f}_{O_c j} = \sum_{j \in \delta^+(O_c)} f_{O_c j} \hat{\rho} = \hat{\rho} \sum_{j \in \delta^+(O_c)} f_{O_c j} = \hat{\rho} F_{\max} = \hat{\rho} u_c^* = d_c$. Thus, this flow satisfies all the demand,
- $\frac{\hat{f}_{ij}}{u_{ij}} = \frac{f_{ij}\hat{\rho}}{u_{ij}} \leq \hat{\rho}$, $\forall (i, j) \in A$,
- $\frac{\hat{f}_{ij}}{u_{ij}} = \frac{f_{ij}\hat{\rho}}{u_{ij}} = \hat{\rho}$, $\forall (i, j)$ such that $(i, j) \in A_c^*$ where $f_{ij} = u_{ij}$.

2. By contradiction. Suppose there exists a flow g_{ij} such that $\frac{g_{ij}}{u_{ij}} < \hat{\rho}$, $\forall (i, j) \in A$. Then, $\sum_{(i,j) \in A_c^*} g_{ij} < \sum_{(i,j) \in A_c^*} u_{ij} \hat{\rho} = \hat{\rho} \sum_{(i,j) \in A_c^*} u_{ij} = \hat{\rho} u_c^* = d_c$. As this flow is not able to satisfy all the demand, we have a contradiction. $\square$

Appendix B. Graphical representation of congested arcs and experienced paths of a specific instance

Graphical representations of congested arcs and experienced paths are presented in Figs. 19 and 20.

(a) $\tau = 0\%$, $\rho_\tau^* = 3.05$ (b) $\tau = 5\%$, $\rho_\tau^* = 2.46$ (c) $\tau = 10\%$, $\rho_\tau^* = 1.79$ (d) $\tau = 15\%$, $\rho_\tau^* = 1.35$ (e) $\tau = 20\%$, $\rho_\tau^* = 1.14$ (f) $\tau = 25\%$, $\rho_\tau^* = 1.02$ (g) $\tau = 30\%$, $\rho_\tau^* = 0.89$ (h) $\tau = 35\%$, $\rho_\tau^* = 0.81$ **Fig. 19.** Congestion for different values of τ .

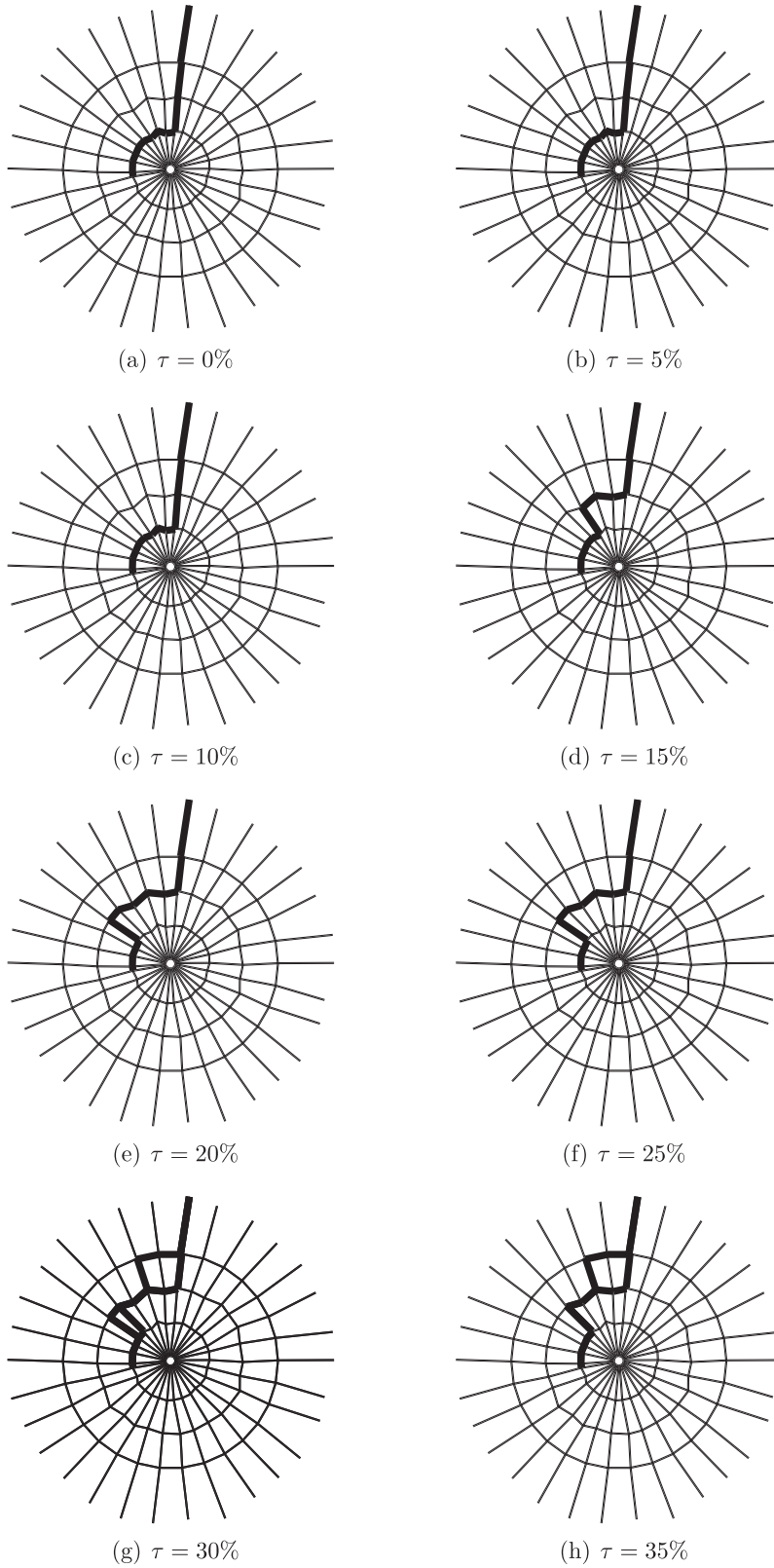


Fig. 20. Selected paths for a single OD pair for different values of τ .

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